DRA-PULSE: Localizing Persona Recoverability Across Stages of a Deep Research Agent

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Abstract

The architecture of a deep research agent—how it routes and re-uses stage artifacts—determines which stages retain recoverable persona signal and which do not. We introduce **DRA-PULSE**, an N-way stage-wise persona attribution diagnostic that recovers the conditioning persona from each intermediate artifact against hard-negative candidates, and apply it to 120 PDR-Bench reports from a four-stage DRA (Gemini 3.6 Flash via `open_deep_research`). Persona recoverability follows a *dip-and-recovery* trajectory: accuracy decreases from Planning (0.808) to Search (0.550), remains low at Compression (0.592), and recovers at Writing (0.800). Persona-to-query alignment thins at query execution (MiniLM cosine: $0.269 \rightarrow 0.212$), explaining the Search dip; the Writing prompt's re-injection of the Planning brief creates the late-stage architectural recovery path. At the report level, 32 runs lose correct attribution between Planning and Search; 24 of these recover by Writing. Replacing the Planning brief with a mismatched persona's brief (write-only ablation, n=15) drops Writing accuracy from 0.933 to 0.467, confirming that brief content—not mere re-injection—drives recovery. The trajectory is robust to identity masking, a second LLM matcher, and equal character-budget controls. We present DRA-PULSE as a process-level diagnostic of how a specific DRA architecture routes persona signal, not a measure of personalization utility or causal influence.

1 Introduction

The architecture of a deep research agent shapes how persona signal propagates across pipeline stages: stages that route rich upstream artifacts directly to their prompts retain recoverability, while stages that transform the conditioning brief into retrieval queries lose it. In `open_deep_research`, the Writing prompt *re-injects the Planning brief* as its top-level instruction, creating an architectural recovery path; query execution at Search, in contrast, strips persona conditioning through progressive semantic thinning. Standard end-to-end personalization metrics cannot detect this internal structure because they observe only the final report. We treat each pipeline stage's artifact as an observable probe and ask how well an independent evaluator can recover the conditioning persona—a process-level diagnostic we call DRA-PULSE. Throughout this paper, *persona signal* refers operationally to artifact information that supports closed-set recovery of the conditioning persona; the term does not imply causal mediation, personalization utility, or preservation of all persona preferences. The trajectory we observe is a characterisation of this specific DRA architecture, not a general property of deep research agents.

Our contributions are:

- **Diagnostic task.** We formalize stage-wise closed-set persona attribution for multi-stage DRA pipelines and implement DRATracer, a deterministic LangGraph listener that records stage artifacts without post-hoc reconstruction.
- **Empirical finding.** On 120 reports we document a report-level *loss-and-recovery* process: 32 Planning→Search losses, of which 24 recover by Writing, with Search as the lowest-recoverability stage.
- **Mechanistic analysis.** Search-view decomposition, Planning-brief re-injection analysis, and shortcut controls bound how architecture and identity cues shape recoverability—without equating Acc@1 with utility.

2 Related Work

2.1 Personalized Generation and Deep Research

Personalization in IR has long used user profiles and behavioral histories for re-ranking (Teevan et al., 2005; Robertson and Zaragoza, 2009). LLM personalization benchmarks such as LaMP (Salemi et al., 2023) and LaMP-QA (Salemi and Zamani, 2025) test conditioning on user histories for classification and long-form QA, with growing surveys of personalization methods for large models (Zhang et al., 2024; Salemi et al., 2024; Richardson et al., 2023). **PDR-Bench** (Liang et al., 2025) targets personalized *deep research*: multi-step, long-form report generation conditioned on rich personas (identity plus actionable preferences). Industry deep-research systems similarly emphasise multi-step search and synthesis (OpenAI, 2025; Google DeepMind, 2024), yet evaluation still centres on final reports. These settings largely lack stage-level diagnostics of *where* persona conditioning remains recoverable inside the pipeline—the gap we address with closed-set N-way stage-wise attribution.

2.2 Authorship Attribution, Probing, and Preference Identification

Classical authorship attribution recovers authors from closed candidate sets via stylometric and neural features (Stamatatos, 2009; Shrestha et al., 2017; Rivera-Soto et al., 2021). Probing classifiers similarly test what attributes are linearly readable from representations (Conneau et al., 2018; Belinkov, 2022), and attribute inference raises dual-use privacy concerns (Pan et al., 2020). Mechanistic work further localises factual associations inside model components (Geva et al., 2021; Meng et al., 2022); we share the *where is the information?* question, but localise recoverability across *pipeline stages and artifacts* rather than weights or activations. Closest to our task, Li and Chaturvedi (2025) propose *comparative personalization* for multi-document summarization as pairwise preference attribution. We differ by (i) N-way simultaneous attribution and (ii) stage-wise localization across a four-stage DRA trajectory, measuring recoverability of the conditioning persona rather than pairwise preference ranking alone.

2.3 Agents, Tool Use, and Process-Level Evaluation

LLM agents combine reasoning and tool use (Yao et al., 2023b; Schick et al., 2023; Wei et al., 2022; Yao et al., 2023a) and are evaluated on interactive suites such as AgentBench (Liu et al., 2024), GAIA (Mialon et al., 2023), and SWE-bench (Jimenez et al., 2024). Multi-hop and retrieval-augmented pipelines further stress intermediate evidence handling (Yang et al., 2018; Lewis et al., 2020; Asai et al., 2024; Press et al., 2023). Deep research agents such as `open_deep_research` (LangChain AI, 2025) (built on LangGraph (LangChain AI, 2024)) elaborate a query into a brief, run parallel search/compress subgraphs, then synthesise a report—so persona-conditioned information must pass through multiple transformations. **PrefDisco** (Li et al., 2025) shows preference accuracy can degrade in reasoning/content-selection layers of multi-step pipelines. Process supervision argues that intermediate steps, not only outcomes, should be scored (Lightman et al., 2023; Uesato et al., 2022); LLM-as-judge protocols likewise evaluate free-form outputs (Zheng et al., 2023). We connect these threads by treating each DRA stage artifact as a closed-set recoverability probe: a process diagnostic of *where* conditioning remains readable, not a substitute for end-to-end personalization quality, for which we also report non-LLM baselines (BM25, embeddings) (Robertson and Zaragoza, 2009; Reimers and Gurevych, 2019).

Stage	Artifact	Content
Planning	a_{plan}	Research brief: scoped topics, angle
Search	a_{search}	Queries + top-3 Serper results each
Compression	a_{compress}	Per-topic synthesis of evidence
Writing	a_{write}	Full long-form research report

Table 1: Four stage artifacts captured per DRA run.

3 Method

3.1 Task Formulation

Let P be a DRA pipeline and $\mathcal{C} = \{p_1, \ldots, p_N\}$ be a candidate persona set ($N = 3$ in our primary experiments). P is conditioned on one **ground-truth persona** $p^* \in \mathcal{C}$ and a research query q, and produces a sequence of **stage artifacts**:

$$\mathcal{A}(q, p^*) = \big(a_{\text{plan}},\ a_{\text{search}},\ a_{\text{compress}},\ a_{\text{write}}\big). \quad (1)$$

The **N-way stage-wise attribution** task is defined as: for each stage artifact $a \in \mathcal{A}$, identify p^* from $\mathcal{C}$:

$$\hat{p} = f\big(a,\ \mathcal{C}\big), \qquad \hat{p} \in \mathcal{C}. \quad (2)$$

Attribution accuracy at stage s is $\text{Acc}_s = \mathbb{E}[\mathbf{1}[\hat{p}_s = p^*]]$, with chance level $1/N$. Stage-wise accuracy profiles localize *where* persona recoverability is high or low: a flat profile suggests little stage-specific change; a peak at a_{plan} that falls at Search and recovers at a_{write} localises a dip-and-recovery pattern in closed-set discriminability.

3.2 DRA Pipeline and Artifact Schema

We use **open_deep_research** (LangChain AI, 2025) as the DRA backbone, instantiated with Gemini 3.6 Flash. The pipeline implements a four-stage LangGraph graph; Table 1 summarizes each stage's artifact.

DRATracer is our LangGraph v2 event listener that captures all four artifacts deterministically from a single pipeline run (implementation artifact; public release planned with the camera-ready version if accepted). The key challenge is that three researcher subgraphs run concurrently under the supervisor; DRATracer resolves per-artifact topic identity using parent run-ID correlation across `on_chain_start` and `on_chain_end` events. All artifacts are serialized to a versioned JSON schema (`dra_trace_v2`) with SHA-256 prompt hashes and a query/source ledger for reconciliation.

3.3 Attribution Protocol

LLM Matcher. We use Upstage Solar Pro (API model ID: `solar-pro`) as the primary attribution model (temperature $= 0$, structured output). Given artifact text a and persona set $\mathcal{C}$ (order-shuffled to prevent position bias), the matcher returns a single predicted

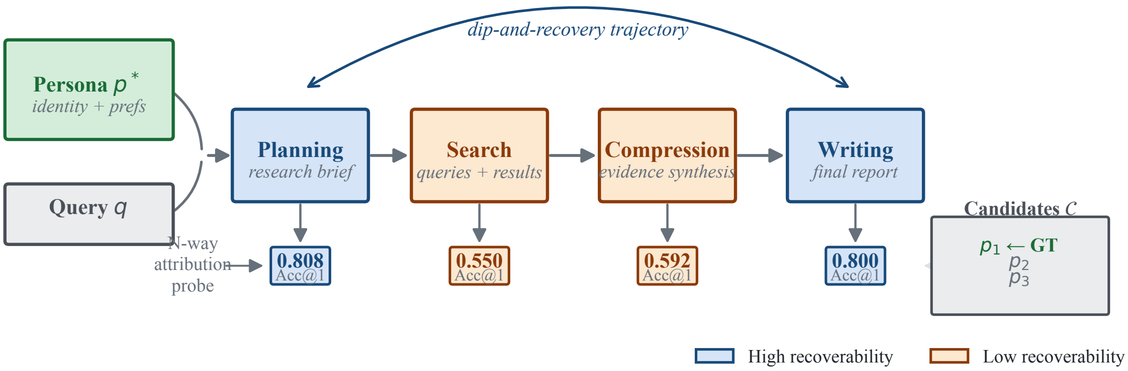

Figure 1: DRA-PULSE framework. A user persona p^* and query q condition a four-stage DRA pipeline (Planning, Search, Compression, Writing). At each stage, an independent N-way attribution probe identifies p^* from a hard-negative candidate set $\mathcal{C}$ using only the stage artifact. Blue stages (Planning, Writing) have high persona recoverability; orange stages (Search, Compression) are low. The dip-and-recovery trajectory (§5) is the core empirical finding.

persona $\hat{p}$. Candidate order is deterministically shuffled per run and stage using a SHA-256-derived seed. To bound matcher context consistently, serialized artifacts are truncated to 4,000 characters for Planning, 3,500 for Search, 6,000 for Compression, and 8,000 for Writing; over-length Writing artifacts retain both the opening and tail of the report.

Non-LLM Baselines. We include three baselines: (1) *Random*, uniform over $\mathcal{C}$ (expected accuracy $= 1/N$); (2) *BM25* (Robertson and Zaragoza, 2009), Okapi BM25 similarity between artifact and persona text; (3) *Embedding* (Reimers and Gurevych, 2019), cosine similarity via a sentence encoder.

Hard Negatives. We remove identity-field tokens—name, age, occupation, education, residence, and family—from each persona and compute Jaccard similarity over the remaining *actionable-preference* tokens (interests, goals, decision style, and related preference text). For each ground-truth persona, the two most similar same-domain profiles are selected as distractors ($\geq$ 0.2 threshold; deterministic nearest-fallback if fewer than two exceed it). This makes chance-level performance a meaningful baseline and selects distractors by preference content rather than demographic identity. Candidate-set-size sensitivity uses prefixes of the same ranking (top-1 for $N = 2$; top-4 for $N = 5$).

3.4 Shortcut Controls

High attribution accuracy could arise from **lexical leakage**—name tokens, demographic phrases, or verbatim persona text appearing in an artifact—rather than broader persona-conditioned content selection. We report one post-processing masking control and one generation-time shuffled-actionable intervention (Table 2); two additional generation-time ablations were attempted but did not pass the pre-specified completeness gate (Appendix G).

Condition	Manipulation	Tests
Full (baseline)	Unmodified persona	Upper bound
Identifier-masked	NER + all-candidate identity-phrase masking	Surface mentions?
Shuffled-actionable	Donor actionable prefs, shell identity	Actionable vs. identity?

Table 2: Shortcut control conditions executed in this paper. Identifier masking is a post-processing pass on all four existing stage artifacts; shuffled-actionable is a 30-report two-seed generation-time intervention. Two additional generation-time conditions (actionable-only, identity-only) were attempted but did not pass the pre-specified completeness gate (Appendix G).

4 Experimental Setup

4.1 Datasets

PDR-Bench (Primary). We use the English split of PDR-Bench (Liang et al., 2025), which provides 25 user personas and 250 query-persona pairs across ten domains (Education, Career, Health, Travel, Finance, Creative, Shopping, Real Estate, Law, and Parenting). We apply domain-stratified largest-remainder sampling to select 20 confirmatory query groups; eligibility filtering removes tasks with fewer than three personas that have at least one actionable token after identity masking. Hard-negative candidate sets are constructed as described in §3. The confirmatory run uses two seeds ($s \in \{0, 1\}$), yielding **120 reports** (20 groups $\times$ 3 personas × 2 seeds). A 5-group development split (15 reports) serves as an engineering gate and is excluded from all research-question analyses.

4.2 Generation Stack

All reports are generated with open_deep_research (LangChain AI, 2025) using Gemini 3.6 Flash for all pipeline roles. Search is provided by a Serper adapter with a report-level execution cap of 7 attempted web calls (at most 2 per researcher invocation), top-3 organic results each, max 15 unique URLs, and a global hard stop at 2,400 attempted queries. The planner may *request* more candidate queries across parallel research topics; rejected proposals are logged but only successful executions (`status=ok`) enter the Search artifact used for attribution. Generation seed is fixed via `generation_config.seed`; other hyperparameters use model defaults. Every artifact records a full `execution_config` block including model IDs, seeds, and SHA-256 hashes for reproducibility. The primary analysis includes all schema-valid reports. We separately report a sensitivity analysis restricted to runs with no completeness errors and to runs satisfying all pre-defined success criteria.

Identifier-masked variants are produced by post-processing existing artifacts at zero additional generation cost.

4.3 Evaluation Metrics

Attribution Accuracy (Acc@1). Fraction of reports for which the matcher's top-1 prediction equals p^*. Reported per stage and macro-averaged across seeds. Chance level: $1/N = 0.33$.

Stage Trajectory. The four-tuple $(\text{Acc}_{\text{plan}}, \text{Acc}_{\text{search}}, \text{Acc}_{\text{compress}}, \text{Acc}_{\text{write}})$ plotted over pipeline depth. A rising trajectory implies increasing recoverability with pipeline depth; a falling then rising pattern is the dip-and-recovery regime we study.

Seed Variance. Std. dev. of Acc@1 across seeds 0 and 1 as a fragility indicator.

5 Results

5.1 Aggregate Stage Trajectory

Table 3 reports the controlled two-seed comparison between Solar Pro and three non-LLM baselines. Solar is strongest overall, but its advantage over BM25 is stage-dependent (Figure 4).

Dip-and-recovery. Across both seeds, recoverability drops from Planning (0.808) to Search (0.550, $\Delta = -0.258$, 95% CI $[-0.333, -0.175]$), changes little at Compression (0.592, $\Delta = +0.042$, $[-0.025, 0.108]$), and recovers at Writing (0.800, $\Delta = +0.208$, $[0.142, 0.283]$). Planning and Writing are nearly equal in the aggregate. The trajectory appears in both generation seeds; the largest seed difference is 0.050 at Compression, and every seed-difference interval includes zero.

Stage	Solar	BM25	Embed	Rnd
Planning	**0.808**	0.633	0.442	0.292
Search	0.550	**0.567**	0.292	0.367
Compression	**0.592**	0.567	0.375	0.383
Writing	**0.800**	0.575	0.475	0.250
Macro avg.	**0.688**	0.585	0.396	0.323
Chance		0.333		

Table 3: N-way stage-wise attribution Acc@1 on PDR-Bench confirmatory split (seeds 0 and 1, $n = 120$, $N = 3$ hard-negative candidates). Solar = Solar Pro (LLM matcher); BM25 = Okapi BM25; Embed = all-MiniLM-L6-v2 cosine; Rnd = random.

Transition	Lost	Gained	Net
Planning → Search	32	1	−31
Search → Compression	8	13	+5
Compression → Writing	6	31	+25

Table 4: Adjacent-stage lost/gained counts (n=120).

5.2 Report-Level Loss and Recovery

Table 4 and Figure 2 decompose the aggregate curve for the *same* reports. Planning→Search is almost purely lossy: 65 stay correct, **32** lose correctness, only **1** gains, and 22 stay wrong. Compression has 13 gains vs. 8 losses (net unresolved at aggregate level); Writing has 31 gains vs. 6 losses. Of the 32 Plan-correct/Search-wrong reports, 24 (75%) recover by Writing (Figure 3); 8 do not (Appendix F). More broadly, 33 of 54 Search errors (61%) are correct at Writing. Exploratorily, the stage ordering appears in nine of ten domains; Creative is the only exception, while Search reaches chance in Career, Finance, and Real Estate (Appendix D).

5.3 Robustness Checks

Beyond lexical matching. The Solar–BM25 gap varies by stage. Across the paired two-seed reports, Solar exceeds BM25 by 0.175 at Planning (task-bootstrap 95% CI $[0.083, 0.258]$) and 0.225 at Writing ($[0.142, 0.308]$). At Search, BM25 is 0.017 higher than Solar, but the paired difference includes zero ($[-0.125, 0.092]$); the Compression difference is similarly unresolved ($+0.025$, $[-0.058, 0.117]$). Thus, the LLM evaluator's advantage is concentrated at Planning and Writing, while the low Search score is shared with a strong lexical baseline. Critically, both non-LLM baselines independently reproduce the Planning-to-Search dip: BM25 drops from 0.633 to 0.567, and Embedding from 0.442 to 0.292. Because these baselines involve no LLM reasoning, the dip pattern reflects artifact-level properties rather than an evaluator-specific bias. The both-evaluator-wrong rate provides a further evaluator-agnostic measure: at Search (seed 0), Solar and GPT-5.4-nano both err simultaneously on

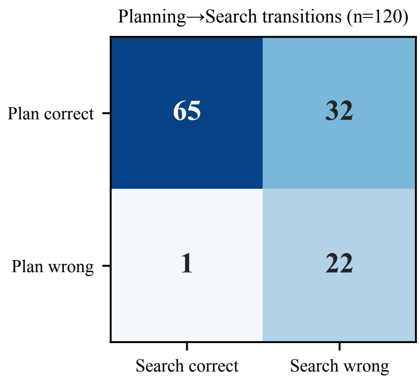

Figure 2: Planning→Search correctness counts (n=120). Almost all change is Plan-correct → Search-wrong (32), not recovery from Plan-wrong (1).

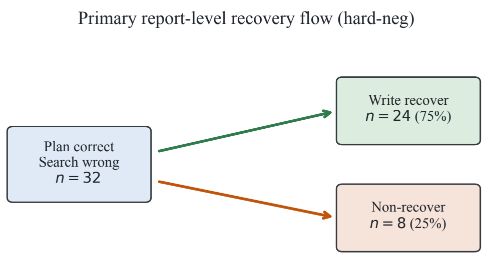

Figure 3: Report-level recovery story for the primary hard-negative protocol: among 32 Plan-correct / Search-wrong reports, 24 (75%) are again correct at Writing and 8 (25%) never recover.

40% of decisions (24 of 60), versus 12% at Writing (7 of 60). This 3.3-fold difference in concurrent evaluator failure confirms that Search artifacts contain less recoverable persona evidence regardless of the evaluator's reasoning capacity.

Candidate-construction sensitivity. Because per-ground-truth hard negatives are selected around the target persona, the candidate set can contain a structural prior. An artifact-free heuristic that selects the candidate with the highest mean pairwise actionable-token Jaccard identifies 27 of 60 construction centers (0.450; task-cluster bootstrap 95% CI $[0.333, 0.567]$). We therefore treat *absolute* hard-negative Acc@1 with caution and treat *paired stage transitions* as the primary scientific target. Importantly, the same report keeps one candidate set across all stages, so a static prior cannot alone generate within-report stage changes. Table 5 summarises protocol and quality subsets. Symmetric task-shared candidates lower absolute Acc (0.733/0.542/0.608/0.708) but preserve Planning→Search (-0.192, CI excludes 0) and late recovery. Splitting by the candidate-only heuristic: *prior-correct* (n=54) and *prior-wrong* (n=66) both keep the dip-and-recovery shape, so the trajectory is not confined to prior-solvable reports. A strict-quality subset (n=90; all pre-defined success criteria) yields

Protocol / subset	n	Plan	Search	Comp.	Write
Per-GT hard-neg	120	0.808	0.550	0.592	0.800
Symmetric task-shared	120	0.733	0.542	0.608	0.708
Prior-correct subset	54	0.907	0.593	0.593	0.833
Prior-wrong subset	66	0.727	0.515	0.591	0.773
Strict-quality subset	90	0.822	0.567	0.611	0.811

Table 5: Stage Acc@1 under candidate protocols, candidate-only-prior splits, and a strict-quality inclusion filter. Dip-and-recovery holds in every row.

Stage	Full	Masked	Δ
Planning	0.808	0.725	-0.083
Search	0.550	0.558	$+0.008$
Compression	0.592	0.575	-0.017
Writing	0.800	0.783	-0.017

Table 6: Solar Acc@1 before and after candidate-derived identity masking. Only Planning changes reliably: $\Delta = -0.083$, task-cluster bootstrap 95% CI $[-0.150, -0.025]$. Intervals for the other three deltas include zero.

0.822/0.567/0.611/0.811, ruling out Search dip as a pure completeness/ledger artefact. Candidate construction affects absolute levels; it does not create the stage ordering.

Protocol-matched shortcut controls. We mask both spaCy PERSON/ORG/GPE/NORP/FAC entities and exact identity-field phrases derived from all three candidate profiles, including names, ages, occupations, education, residence, and family fields. Applying phrases from all candidates avoids using the ground-truth profile alone to define the transformation. The masked and full conditions have identical candidate sets and SHA-256 orders for all 480 paired decisions. Results are shown in Table 6.

The masked trajectory still drops from Planning to Search (-0.167, $[-0.258, -0.067]$) and recovers from Compression to Writing ($+0.208$, $[0.100, 0.308]$). Thus, direct identity phrases explain part of Planning recoverability but do not explain the late-stage recovery.

Cross-matcher replication. On the frozen seed-0 cohort, GPT-5.4-nano produces 0.750/0.467/0.617/0.850 across Planning, Search, Compression, and Writing. Its Planning-to-Search drop is -0.283 (task-cluster bootstrap 95% CI $[-0.433, -0.133]$), followed by a Compression-to-Writing gain of $+0.233$ ($[0.083, 0.383]$). Solar and GPT agree on 188 of 240 decisions (0.783; Cohen's $\kappa = 0.763$). Stage-wise κ mirrors the primary trajectory: highest at Writing (0.855) and Planning (0.837), and lower at Search (0.680) and Compression (0.673)—the two stages with the lowest attribution accuracy. On seed 1, GPT-5.4-nano produces 0.833/0.583/0.600/0.733, agreeing with Solar on 76.7% of decisions ($\kappa = 0.745$); the dip-and-recovery

Stage	Solar	GPT-nano	Agree	κ
Planning	0.800	0.750	0.850	0.837
Search	0.533	0.467	0.717	0.680
Compression	0.567	0.617	0.700	0.673
Writing	0.817	0.850	0.867	0.855
All	0.679	0.671	0.783	0.763

Table 7: Per-stage cross-matcher comparison (seed 0, $n = 60$). Solar = Solar Pro; GPT-nano = GPT-5.4-nano-2026-03-17. Agreement and κ are highest where attribution accuracy is highest (Planning, Writing) and lowest where it is lowest (Search, Compression).

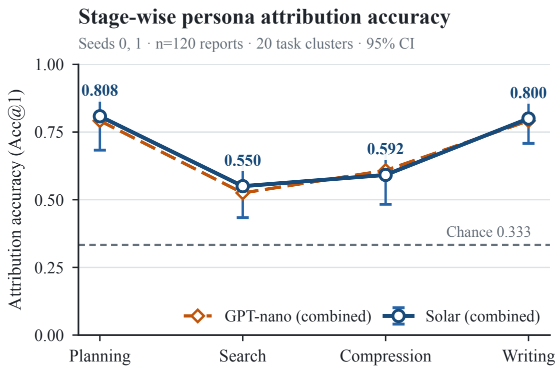

Figure 4: Stage-wise attribution trajectory. Blue (solid): Solar Pro combined ($n = 120$, two seeds) with 95% task-cluster bootstrap intervals. Orange (dashed): GPT-5.4-nano combined mean of the same two seeds. Both evaluators show the same dip-and-recovery shape.

shape is consistent across evaluators and seeds. Table 7 gives per-stage detail for seed 0.

Quality and domain sensitivity (exploratory). Restricting analysis to the 90 reports satisfying all pre-defined success criteria yields 0.822/0.567/0.611/0.811; excluding completeness errors yields 0.824/0.571/0.615/0.813 ($n = 91$). Both retain dip-and-recovery. Domain patterns ($n = 12$ per cell) are exploratory and deferred to Appendix D.

Candidate-set-size sensitivity. The trajectory also holds for $N{=}2$ and $N{=}5$, with nearly unchanged chance-normalized Planning accuracy (Table 16, Appendix H).

6 Analysis

6.1 Why Does Search Accuracy Dip?

The Search artifact combines executed queries with retrieved titles/snippets and is the lowest-recoverability stage (32 Plan→Search losses vs. 1 gain). Table 8: queries-only Acc@1 is 0.600 vs. 0.533 snippets-only and 0.550 full Search (queries−full $+0.050$, CI $[0.000, 0.108]$, same in both seeds). Cross-seed within-

View	Seed 0	Seed 1	Combined
Full Search	0.533	0.567	0.550
Queries only	0.583	0.617	0.600
Snippets only	0.483	0.583	0.533

Table 8: Search Acc@1 by artifact view and generation seed ($n = 60$ per seed; Solar Pro matcher; 240 total decisions, SHA-256 order audit 240/240 pass). The queries-only advantage over full Search is $+0.050$ in each seed independently.

persona URL overlap (0.042) is near cross-persona overlap on the same query (0.034), so retrieved sets show little persona-specific discriminability by URL metrics.

Where persona alignment thins. We probe lexical and embedding-based alignment from persona profiles into Planning briefs and executed Search queries (Figure 5; protocol in Appendix L). Lexically, mean actionable-token retention is already low in briefs (0.068) and near-zero in executed queries (0.008). Under an embedding-based similarity probe (MiniLM cosine), mean persona–brief alignment is 0.269 and falls to 0.212 for persona–query (mean drop 0.057, task-cluster bootstrap 95% CI $[0.042, 0.075]$); brief–query remains higher (0.493), so queries still track the research topic while losing persona conditioning. Persona–query similarity is also higher on Search-correct than Search-wrong runs (0.236 vs. 0.182; difference 0.055, CI $[0.019, 0.088]$).

Which persona features disappear? Section-level token retention (Figure 6) shows the same compression across profile buckets: identity (0.100→0.020), interests (0.081→0.010), goals/constraints (0.075→0.008), and decision style (0.020→0.002). The Planning→Search loss is therefore not explained by a single missing identity string: multiple persona sections thin when the agent reformulates the brief into retrieval queries.

Illustrative compression. On `pilot_task12_User5_seed0`, Planning correctly attributes an engineering graduate seeking a six-month emotion-regulation curriculum. Search collapses to book-title / ISBN lookups ("Feeling Good", "Mind Over Mood"), persona–query MiniLM similarity drops to 0.115, and attribution flips to User2; Writing recovers User5 after brief re-injection. A second recovery case (AI-education startup) is in Appendix E.

Structural origin of Writing recovery. The Writing prompt re-injects the Planning research brief as its top-level instruction, creating an architectural recoverability path. Table 9: $P(\text{Write}\checkmark \mid \text{Plan}\checkmark) = 0.887$ vs. 0.435 when Planning attribution is incorrect. The conditional gap is consistent with dependence on the Planning artifact; the 43.5% rate when Planning is wrong

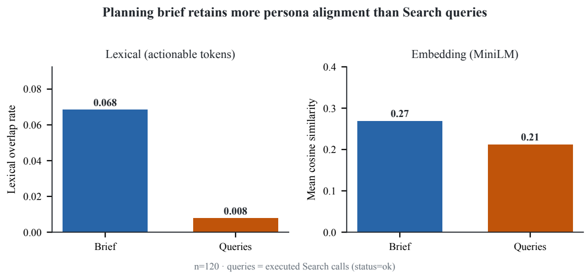

Figure 5: Persona alignment of Planning briefs vs. executed Search queries (n=120). Left: lexical overlap of actionable tokens (0.068→0.008). Right: MiniLM cosine similarity (0.269→0.212). Both drop at query execution; residual embedding alignment separates Search-correct vs. wrong runs (0.236 vs. 0.182).

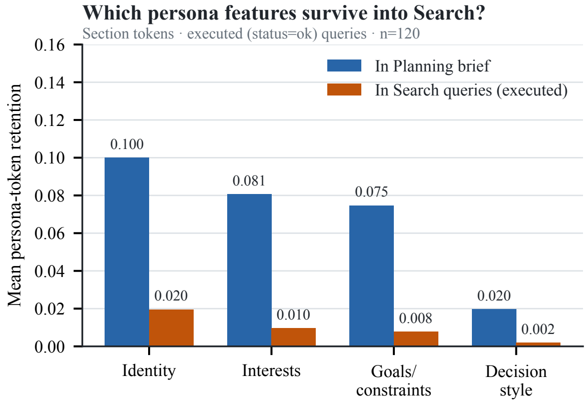

Figure 6: Section-level persona-token retention in Planning briefs vs. executed Search queries (n=120). Identity, interests, and goals/constraints all drop; decision-style language is already sparse in briefs. Field-to-bucket mapping in Appendix L.

shows Writing recoverability is not deterministically tied to the Planning attribution outcome.

Write-only brief interventions. We freeze Plan/Search/Compress and rewrite finals (Figure 7). On a 15-report slice, injecting another persona's brief drops Write Acc@1 from 0.933 to 0.467 ($\Delta = -0.467$; CI $[-0.667, -0.200]$). Under mismatched briefs, predictions are **GT 7 / Donor 1 / Other 7**: the wrong brief disrupts target recoverability *without* simply transferring attribution to the donor. Removing the brief entirely on 30 seed-0 runs yields a smaller drop to 0.833 ($\Delta = -0.067$; CI $[-0.167, +0.067]$), with the CI including zero. Brief content—not mere presence—is therefore the interventionally relevant factor for Writing recoverability.

Attribution vs. personalization utility. A GPT-4o-mini exploratory judge on 120 reports yields mean 4.01/3.65 (content/presentation); Write-correct vs. wrong differs by only 0.10 (r=0.15; CI includes 0). Acc@1 is not a utility substitute.

	Write ✓	**Write ×**
Plan ✓	86	11
Plan ×	10	13

Table 9: Planning $\times$ Writing co-occurrence (n=120). Conditional rates: $P(\text{Write}✓ \mid \text{Plan}✓) = 0.887$; $P(\text{Write}✓ \mid \text{Plan wrong}) = 0.435$.

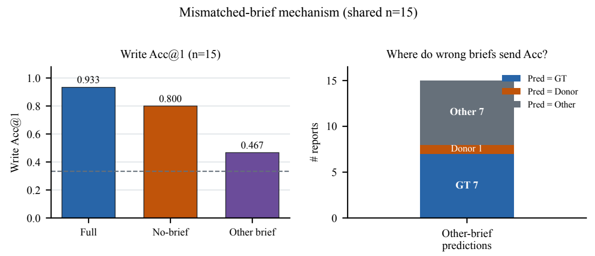

Figure 7: Left: Write Acc@1 under full / no-brief / other-persona brief (n=15). Right: prediction breakdown under other-brief (GT / Donor / Other).

6.2 Shortcut Controls

Identity masking. The strong Planning reduction under masking indicates that early-stage recoverability partly uses direct identity evidence. In contrast, Search, Compression, and Writing change by at most 0.017, and the masked Writing accuracy remains 0.783. This is consistent with later recoverability drawing on evidence beyond exact candidate identity phrases. It does not prove that the remaining evidence corresponds exclusively to actionable preferences: paraphrased identity cues and other lexical shortcuts may remain.

Shuffled-actionable intervention. Across 30 reports (seeds 0–1), we replace each identity shell's actionable preferences with a donor's. Table 10: shell dominates donor at Planning (Shell−Donor +0.433, CI $[0.100, 0.800]$) and Writing (+0.567, $[0.267, 0.867]$); Search is unresolved (+0.133). DRA-PULSE thus detects which conditioning persona remains identifiable, with recoverability anchored more in the identity shell than in transplanted preferences—a metric boundary, not proof that the model ignores preferences.

7 Limitations

Attribution measures whether a matcher can recover the conditioning persona; it does not measure usefulness, factual quality, user satisfaction, or causal influence. MiniLM cosine is an exploratory embedding-based alignment proxy and may miss paraphrased or compositional persona cues; section-level token retention is likewise a surface lexical diagnostic, not a causal feature ablation.

Single-backbone scope. All results come from one DRA implementation (Gemini 3.6 Flash via `open_deep_research`). The Writing-stage brief re-injection is an architectural feature of this specific

Stage	Shell	Donor	Other	Δ	95% CI
Planning	0.633	0.200	0.167	+0.433	[+0.100, +0.800]
Search	0.433	0.300	0.267	+0.133	[−0.133, +0.567]
Compression	0.500	0.233	0.267	+0.267	[0.000, +0.667]
Writing	0.733	0.167	0.100	+0.567	[+0.267, +0.867]

Table 10: Shuffled-actionable Solar prediction rates (n=30). Δ=Shell−Donor; task-cluster bootstrap 95% CI.

pipeline; DRA systems that do not re-inject the planning artifact would likely show a different trajectory shape. A write-only no-brief ablation on 30 seed-0 reports softens the pure tautology reading (Write Acc@1 remains 0.833 without the brief), but does not replace multi-backbone or multi-pipeline replication. The dip-and-recovery pattern as reported here is therefore a characterisation of this architecture, not a general property of DRA pipelines.

Other limitations. The shuffled-actionable control shows recoverability is anchored more in identity-shell cues than transplanted preferences; DRA-PULSE therefore measures recoverability of the identity layer more reliably than of actionable persona content. Default matcher budgets differ by stage (Search 3.5k / Write 8k chars), but BM25, queries-only Search, and equal-budget rematch (seed 0, 3.5k all stages) preserve dip-and-recovery. Matchers lack human attribution validation; seeds share 20 task clusters; hard-negatives carry a candidate-only prior (absolute Acc caution); Search views are post-hoc; domain/recovery cells and utility scores are exploratory. Acc@1 is not causal mediation or utility.

Ethics Statement

Stage-wise persona attribution can be dual-use for profiling system logs. We study synthetic PDR-Bench personas in closed candidate sets only. Recoverability metrics must not be treated as privacy or personalization quality certificates without human review and domain safeguards.

8 Conclusion

We introduced N-way stage-wise persona attribution as a process-level diagnostic of *persona recoverability*, with DRATracer capturing intermediate artifacts. On 120 reports, 32 cases lose attribution from Planning to Search and 24 recover by Writing (trajectory 0.808/0.550/0.592/0.800). The shape holds under lexical baselines, a second matcher, masking, symmetric candidates, and N-sensitivity; brief interventions and utility checks bound re-injection and Acc≠utility. Results characterise one DRA stack, not multi-backbone generality or downstream usefulness.

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A Full Quality Sensitivity

Table 11 reports Solar Acc@1 under four inclusion policies. The dip-and-recovery ordering is preserved under every restriction, confirming that the primary result is not driven by low-quality reports.

Policy	Plan	Search	Comp.	Write	n
All matched	0.808	0.550	0.592	0.800	120
Success criteria met	0.822	0.567	0.611	0.811	90
No completeness err.	0.824	0.571	0.615	0.813	91
No ledger errors	0.807	0.546	0.588	0.798	119

Table 11: Solar Acc@1 under four quality-filtering policies (Solar Pro matcher, per-GT hard-negative candidates). All policies preserve the same stage ordering.

B Per-Seed Baseline Results

Table 12 disaggregates the non-LLM baseline accuracies by generation seed. BM25 exceeds Solar at Search in seed 0 (0.600 vs. 0.533) but not seed 1 (0.533 vs. 0.567), so the small combined advantage is not seed-stable. Random accuracy varies around the 0.333 chance level, as expected.

Seed	Method	Plan	Search	Comp.	Write
0	BM25	0.667	0.600	0.583	0.583
0	Embedding	0.450	0.300	0.383	0.483
0	Random	0.350	0.367	0.400	0.250
1	BM25	0.600	0.533	0.550	0.567
1	Embedding	0.433	0.283	0.367	0.467
1	Random	0.233	0.367	0.367	0.250

Table 12: Non-LLM baseline Acc@1 by generation seed ($n = 60$ per seed; per-GT hard-negative candidates). BM25 leads Solar at Search in seed 0 (0.600 vs. 0.533) but not seed 1 (0.533 vs. 0.567), so the combined BM25 advantage at Search (+0.017) is seed-0 driven and does not replicate.

C Extended Persona Confusion Table

Table 13 lists all ground-truth/prediction pairs with two or more stage-level errors across the 120 reports. User10 appears as the predicted label for eight distinct ground-truth personas, concentrated at Search and Compression.

D Stage Accuracy by Domain

Figure 8 visualises the full domain $\times$ stage accuracy matrix. Table 14 gives the same data numerically.

Table 14 shows Solar Acc@1 across the ten PDR-Bench domains (both seeds, $n = 12$ per domain per stage). The dip-and-recovery trajectory holds in nine of ten domains. The exception is Creative, where Writing accuracy (0.500) does not rise above Search (0.500), suggesting that open-ended creative tasks may be harder to attribute once the planning brief's topic framing is diluted by retrieved content. Education reaches near-ceiling performance (0.917/1.000/0.917/1.000), while Career and Real Estate show the lowest macro-average accuracy (0.500 and 0.458). The Search dip reaches chance level (0.333) in Career, Finance, and Real Estate.

E Qualitative Recovery Trajectories

We highlight two recovered reports (Plan correct, Search wrong, Write correct). Both are illustrative, not statistical claims.

Case A: emotion-regulation curriculum (`pilot_task12_User5_seed0`)

Planning (correct → User5). The brief names the occupation and frames a six-month self-study plan for emotional triggers, CBT/mindfulness resources, and interpersonal regulation— highly persona-specific scope.

Search (wrong → User2). Issued queries collapse to book-title / ISBN lookups (e.g., "Feeling Good" Burns; "Mind Over Mood"; DBT workbook), with little residual occupation or decision-style framing. The matcher flips to User2.

GT	Predicted	Stage	Count
User18	User10	Search	6
User12	User10	Compression	4
User12	User10	Search	4
User17	User10	Search	4
User18	User10	Compression	4
User18	User10	Writing	4
User18	User8	Search	4
User10	User18	Compression	3
User11	User10	Search	3
User14	User13	Search	3
User14	User20	Search	3
User17	User10	Compression	3
User17	User10	Planning	3
User17	User10	Writing	3
User18	User8	Compression	3
User19	User10	Compression	3
User19	User10	Writing	3
User10	User8	Search	2
User11	User10	Compression	2
User12	User10	Planning	2
User13	User10	Search	2
User13	User14	Compression	2
User13	User14	Planning	2
User13	User14	Search	2
User13	User14	Writing	2
User14	User13	Compression	2
User17	User25	Compression	2
User17	User25	Search	2
User17	User25	Writing	2
User18	User10	Planning	2
User19	User10	Search	2
User19	User18	Planning	2
User19	User18	Search	2
User2	User1	Compression	2
User22	User10	Search	2
User5	User2	Search	2
User8	User10	Writing	2
User8	User18	Search	2
User9	User11	Compression	2
User9	User12	Compression	2
User9	User12	Search	2

Table 13: All persona confusion pairs with ≥ 2 stage-level errors across 120 reports (Solar Pro matcher). User10 concentrates misattributions across eight ground-truth personas at Search and Compression.

Domain	Plan	Search	Comp.	Write	Avg
Career	0.667	0.333	0.333	0.667	0.500
Creative	0.750	0.500	0.500	0.500	0.562
Education	0.917	1.000	0.917	1.000	0.958
Finance	0.667	0.333	0.417	0.833	0.562
Health	0.917	0.583	0.667	1.000	0.792
Law	0.833	0.583	0.750	0.833	0.750
Parenting	0.917	0.500	0.750	0.750	0.729
Real Estate	0.500	0.333	0.333	0.667	0.458
Shopping	0.917	0.667	0.667	0.917	0.792
Travel	1.000	0.667	0.583	0.833	0.771

Table 14: Solar Acc@1 by domain across both seeds ($n = 12$ per cell; Solar Pro matcher; per-GT hard-negative candidates). The dip-and-recovery ordering holds in all domains except Creative. Chance accuracy is 0.333.

Compression (wrong → User2). Compressed notes continue to centre catalogued books/courses rather than the engineering-student framing.

Writing (correct → User5). The final report reintroduces an analytical / engineering framing of emotion regulation as a structured information-processing system and recovers User5, consistent with Planning-brief re-injection

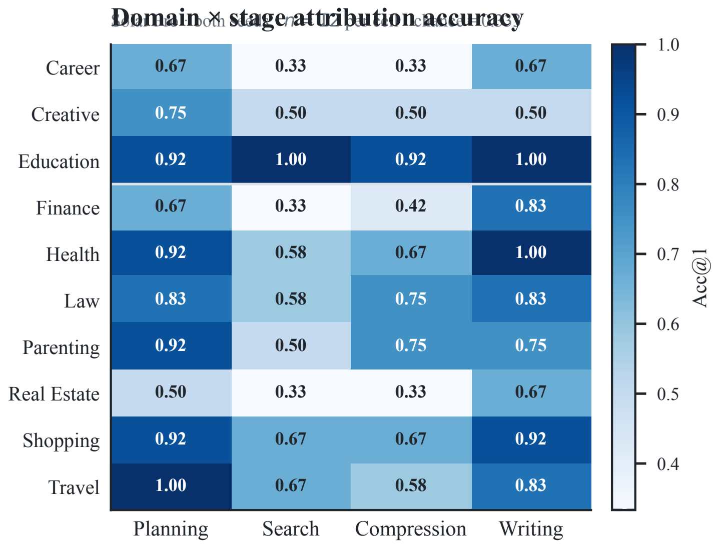

Figure 8: Domain × stage attribution accuracy (Solar Pro, both seeds, $n = 12$ per cell). Colour intensity encodes Acc@1 above the 0.333 chance level (dashed line on colour bar). Education is the only domain where Search reaches ceiling (1.000); Career, Finance, and Real Estate show Search at chance (0.333). Creative is the only domain where Writing (0.500) does not exceed Search.

plus writing-time synthesis.

Case B: AI-education startup (`pilot_task7_User18_seed1`)

Planning (correct → User18). The brief frames launching an online AI education startup for university students, with execution goals around learning efficiency, courses, and integrated tools.

Search (wrong → User8). Queries shift to market/metric retrieval (student AI-tool adoption, LTV/CAC, Chegg/Khanmigo/Gauth case studies). Persona-specific founder constraints thin out; attribution flips to User8.

Writing (correct → User18). The final report re-centres the startup-execution plan and recovers User18.

F Recovery vs. Non-Recovery Report Characteristics

Of the 32 reports that are correct at Planning and wrong at Search, 24 (75.0%) recover to a correct Writing attribution and 8 do not. Table 15 summarises the domain distribution. The 8 non-recovered reports have a lower Compression accuracy (0.250) than the 24 recovered reports (0.375); with n=8 this is descriptive only. Non-recovered reports are spread across six domains; Creative (2/8) and Travel (2/8) are somewhat over-represented relative to their share of all reports. Four of the eight non-recovered reports come from two (task, persona) pairs that fail across both seeds (task17/User17, task29/User18).

Domain	**Recovered** ($n = 24$)	**Not recovered** ($n = 8$)
Career	3	1
Creative	1	2
Finance	4	0
Health	4	0
Law	3	0
Parenting	4	1
Real Estate	1	1
Shopping	2	1
Travel	2	2
Compression Acc	0.375	0.250

Table 15: Domain distribution of the 32 Planning-correct/Search-wrong reports by Writing recovery status. Non-recovered reports have lower Compression accuracy and are modestly concentrated in Creative and Travel.

G EXP-011: Generation-time Ablation Outcomes

We attempted seed-0 generation for two generation-time ablation conditions: *actionable-only* (identity leaf keys removed from the persona conditioning prompt) and *identity-only* (only identity leaf keys retained). Each condition targeted 15 reports (5 task clusters $\times$ 3 personas, seed 0).

Both conditions produced 15/15 schema-valid artifacts. However, each violated the pre-specified completeness gate ($\leq$ 2 reports with missing topic sources), with 3 completeness issues per condition. The actionable-only condition additionally had one query-source ledger mismatch. Consequently, `generation_gate_passed = false` for both conditions, and seed-1 generation was not approved under the pre-registered gate protocol (DEC-006).

The Solar matcher was not run on these artifacts. Because the gate failure disqualifies these results from confirmatory use, we do not report attribution accuracy for the ablation conditions. The primary shortcut evidence in this paper relies instead on the post-hoc identifier-masking control (Table 6) and the generation-time shuffled-actionable intervention (Table 10), both of which passed their pre-specified quality criteria.

H Candidate-Set-Size Sensitivity

Table 16 reproduces the full sensitivity results for all three candidate-set sizes (seeds 0 and 1, $n = 120$ per row; Solar Pro matcher). The Planning→Search drop is -0.209 at $N = 2$, -0.258 at $N = 3$, and -0.287 at $N = 5$, growing monotonically with N. Writing accuracy exceeds Search by $+0.142$ ($N = 2$), $+0.250$ ($N = 3$), and $+0.246$ ($N = 5$). All stages exceed their respective chance levels at all three settings, confirming that the dip-and-recovery pattern is robust to the number of hard-negative candidates.

N	**Chance**	**Plan**	**Search**	**Comp.**	**Write**	$\Delta_{\mathrm{P}\to\mathrm{S}}$
2	0.500	0.842	0.633	0.625	0.775	-0.209
3	0.333	0.808	0.550	0.592	0.800	-0.258
5	0.200	0.758	0.471	0.525	0.717	-0.287

Table 16: Candidate-set-size sensitivity: Solar Acc@1 by N (seeds 0 and 1, $n = 120$ per row). Candidates selected by same-domain actionable-token Jaccard ranking. The trajectory shape is stable across all sizes.

I Matcher Prompt Template

The LLM matcher receives a fixed system prompt and a stage-specific user prompt constructed as follows.

System prompt.

> You are an expert document analyst specializing in user profiling. Given a document excerpt from a research report and a set of candidate user profiles, identify which user this document was most likely produced for. Focus on content priorities, framing, vocabulary, and topic angles that reflect the user's background, goals, and preferences. Call `submit_attribution` with your answer.

User prompt structure.

1. A stage-specific header (*Research Planning Brief*, *Search Queries and Retrieved Sources*, *Compressed Research Summaries*, or *Final Research Report (excerpt)*) followed by the artifact text, truncated to the stage character limit (Planning 4,000; Search 3,500; Compression 6,000; Writing 8,000 with opening and tail preserved).

2. A **Candidate User Profiles** section listing each candidate as **Candidate A / B / C** with the full persona text.

3. A closing instruction: *"Which candidate profile (A, B, or C) best matches the perspective, priorities, and content framing of the document above? Call* `submit_attribution`*."*

Candidate order is deterministically shuffled per run and stage using a SHA-256-derived seed, ensuring position bias cannot produce a systematic accuracy advantage for any fixed candidate slot. The matcher is called with temperature 0 and a structured `submit_attribution` tool that enforces a single-label response.

J DRATracer Artifact Schema

DRATracer is a LangGraph v2 event listener that captures all four pipeline artifacts deterministically from a single pipeline run and writes them as a `dra_trace_v2` JSON object. Table 17 summarises the top-level fields.

Field	Contents
`schema_version`	Schema identifier (`dra_trace_v2`)
`run_id`	Unique run identifier
`research_brief`	Planning stage artifact (text)
`research_topics`	Ordered topic list from the planning stage
`search_trace`	List of per-topic search records: queries issued, sources selected with titles and snippets
`compressed_research`	Per-topic evidence synthesis (Compression stage)
`final_report`	Full Writing stage artifact (text)
`execution_config`	Generation seed, model IDs, prompt SHA-256, manifest version, task/persona IDs
`token_ledger`	Input/output/total tokens, query counts (attempted, successful, failed), source counts
`capture_completeness`	Boolean flags for each required artifact field

Table 17: Top-level fields of the `dra_trace_v2` artifact schema produced by DRATracer.

K Candidate Construction Protocol

For each ground-truth persona p^* in a task cluster, we select two distractor personas from the same domain as follows.

1. **Actionable-token extraction.** Each persona profile is tokenised (lowercased, whitespace-split) and identity-field tokens (name, age, occupation, education, residence, family) are removed, leaving the *actionable token set*.
2. **Jaccard similarity.** For each candidate persona $p_i \neq p^*$ in the same domain, we compute the Jaccard similarity of the actionable token sets: $J(p^*, p_i) = |A^* \cap A_i|/|A^* \cup A_i|$.
3. **Hard-negative selection.** The two personas with the highest $J(p^*, \cdot)$ among those satisfying $J \geq 0.2$ are chosen as distractors. If fewer than two satisfy this threshold, the nearest remaining same-domain personas are used as fallback (tie-broken by ascending user ID).
4. **Frozen per-GT set.** Each ground-truth persona has its own two-distractor set. The resulting $N = 3$ candidate set is fixed in the manifest and shared across all four stages and both generation seeds for that persona. Candidate presentation order is independently shuffled per run and stage using a SHA-256-derived seed.

An artifact-free heuristic that picks the candidate with the highest mean pairwise actionable Jaccard scores 0.450 (bootstrap 95% CI $[0.333, 0.567]$), establishing a structural prior above the 0.333 chance level. This motivates our additional task-shared sensitivity analysis (Section 5).

L Persona Alignment Measurement Protocol

This appendix specifies the surface probes used in §6 (Figures 5 and 6).

Query text. All Search-side measurements use only executed Serper calls with `status=ok` (mean 5.2 per report under the 7-attempt cap). Planner-requested but rejected queries are excluded so that the probe matches the Search artifact seen by the matcher.

Tokenisation. Text is lowercased and split on `[a-z0-9]+`. Tokens of length ≤ 2 and a fixed English stopword list are removed. No stemming or lemmatisation is applied.

Actionable lexical retention. Let A_p be the actionable token set of persona p after removing identity leaf fields (same definition as candidate construction). For artifact text t with token set A_t,

$$\mathrm{Ret}(p, t) = \frac{|A_p \cap A_t|}{|A_p|}.$$

We report means of Ret for briefs and for joined executed queries.

Section-level feature buckets. PDR-Bench profile subtrees are mapped to four buckets without overlap of section paths:

- **Identity:** `Basic Attributes.Identity Characteristics`, `Family Status`, `Long-term Spatial Characteristics`.

- **Decision style:** `Behavioral Characteristics.Personality Traits`, `Environment.Time`.
- **Interests:** `Preferences and Interests`, `Online Usage Habits`, `Offline Long-term Behavior`.
- **Goals/constraints:** `Health Status`, `Financial Information`.

For each bucket B with token set A_B, retention is $|A_B \cap A_t|/|A_B|$. A token string may appear in multiple buckets if it occurs in multiple fields; paths themselves are disjoint.

Embedding-based alignment. We encode (i) the full flattened persona profile, (ii) the Planning brief, and (iii) the concatenation of executed queries with `all-MiniLM-L6-v2` and report cosine similarity. Contrasts use task-cluster bootstrap 95% intervals (2,000 resamples over the 20 task IDs).